Eleventh Section. Ideals in quadratic fields.

§ 90. Discriminant of the quadratic field.

A quadratic field arises when one adjoins a square root $\sqrt{d}$ to the field of rational numbers; here d may be assumed to be an integer without a square divisor. The field is real or imaginary according as d is positive or negative. If one denotes the field of rational numbers, the absolute domain of rationality, by $\mathfrak{R}$, and by $\mathfrak{R}(x, x', \ldots)$ the field which arises by adjoining $x, x', \ldots$ to $\mathfrak{R}$, then the quadratic field Ω is to be denoted by

$$\Omega = \mathfrak{R}(\sqrt{d}). \tag{1}$$

Every number of the field Ω can be put in the form

$$\omega = \frac{x + y\sqrt{d}}{2}, \tag{2}$$

where x, y are rational integral or fractional numbers. The number conjugate to ω,

$$\omega' = \frac{x - y\sqrt{d}}{2}, \tag{3}$$

is likewise contained in the field Ω, and consequently Ω is a normal field.

In order that ω be an integer it is necessary that

$$\omega + \omega' = x, \qquad (\omega - \omega')^2 = y^2 d$$

be rational integers; and since d is to have no square divisor, x and y must be integers. But in order that ω really be an integer it is also still necessary that the norm $\frac{1}{4}(x^2 - y^2 d)$ be a rational integer, that is, it must be

$$x^2 - y^2 d \equiv 0 \pmod 4. \tag{4}$$

If $d \equiv 2$ or $\equiv 3 \pmod 4$, then this condition can be fulfilled only when x and y are even, and if therefore we replace x, y by $2x, 2y$, there follows:

1. If $d \equiv 2, 3 \pmod 4$, then all and only those numbers of the field Ω are integral which are contained in the form

$$x + y\sqrt{d}$$

with rational integers x, y.

If, however, $d \equiv 1 \pmod 4$, then (3) is satisfied when x and y are both even or both odd, hence when x has the form $2x_1 - y$. If one then again replaces x_1 by x, one obtains:

2. If $d \equiv 1 \pmod 4$, then all and only those numbers in Ω are integral which are contained in the form

$$x + y\frac{-1 + \sqrt{d}}{2}$$

with rational integers x, y.

Thus in case 1, $(1, \sqrt{d})$, and in case 2, $\left(1, \frac{-1+\sqrt{d}}{2}\right)$, is a minimal basis of the field Ω. The fundamental number, or discriminant, of the field Ω is therefore, in case 1,

$$\Delta = \begin{vmatrix} 1 & \sqrt{d} \\ 1 & -\sqrt{d} \end{vmatrix}^2 = 4d, \tag{5}$$

and in case 2,

$$\Delta = \begin{vmatrix} 1 & \dfrac{-1 + \sqrt{d}}{2} \\ 1 & \dfrac{-1 - \sqrt{d}}{2} \end{vmatrix}^2 = d. \tag{6}$$

In both cases, therefore, the integral number ω can be put in the form

$$\omega = \frac{x + y\sqrt{\Delta}}{2}, \tag{7}$$

with the condition that

$$4N(\omega) = x^2 - \Delta y^2 \equiv 0 \pmod 4 \tag{8}$$

be satisfied.

The two forms of the minimal basis which we have obtained, namely $(1, \sqrt{d})$ and $\left(1, \frac{-1+\sqrt{d}}{2}\right)$, are therefore

$$(1, \theta) = \left(1, \frac{1}{2}\sqrt{\Delta}\right), \quad \Delta \equiv 0, \qquad = \left(1, \frac{-1+\sqrt{\Delta}}{2}\right), \quad \Delta \equiv 1 \pmod 4. \tag{9}$$

Here the sign of $\sqrt{\Delta}$ may be determined in any arbitrary way, but in the same consideration it is then to be held fixed. We shall assume once for all:

3. If Δ is positive, then $\sqrt{\Delta}$ is likewise to be positive; and if Δ is negative, then $-i\sqrt{\Delta}$ is to be positive, or $\sqrt{\Delta}$ is to be positive imaginary.

From (1), (2) there still follows:

4. The fundamental number of a quadratic field is always $\equiv 0$ or $\equiv 1$ modulo 4 and, apart from 4, has no square divisor. It is therefore a discriminant, indeed a fundamental discriminant.